

WEEK-4: DAY-17 Hands-On Keerthivasan R

Level-1 Problem 1: Simple Calculator Using Methods

Scenario:

A small retail shop wants a simple calculator application to perform addition and subtraction operations using reusable methods.

Requirements:

1. Create a class named Calculator.
2. Create methods Add(int a, int b) and Subtract(int a, int b).
3. Each method should return the result.
4. In Main(), create an object and call the methods.
5. Display the output.

Technical Constraints:

1. Use method parameters and return types properly.
2. Use appropriate access modifiers.
3. No global variables allowed.

Expectations:

Proper method definition, object creation, and method invocation.

Learning Outcome:

Understand functions, parameters, return types, classes, and objects.

Sample Input: 10 5

Sample Output: Addition = 15, Subtraction = 5

NORMAL:

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDAY17
{
    internal class CalculatorApp
```

```

{
    public int Add(int a, int b)
    {
        return a + b;
    }

    public int Subtract(int a, int b)
    {
        return a - b;
    }
}

class Program2
{
    static void Main(string[] args)
    {
        CalculatorApp calc = new CalculatorApp();

        Console.Write("Enter first number: ");
        int num1 = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter second number: ");
        int num2 = Convert.ToInt32(Console.ReadLine());

        int addResult = calc.Add(num1, num2);
        int subResult = calc.Subtract(num1, num2);

        Console.WriteLine("Addition = " + addResult);
        Console.WriteLine("Subtraction = " + subResult);
    }
}

```

OUTPUT:

```

Enter First Number: 20
Enter Second Number: 10
Addition = 30
Subtraction = 10
|

```

ENHANCED:

```
using System;

namespace HandsOnDAY17
{
    internal class Calculator
    {
        // Private data members
        private int num1;
        private int num2;

        // Public method to set values
        public void SetValues(int a, int b)
        {
            num1 = a;
            num2 = b;
        }

        // Method for addition
        public int Add()
        {
            return num1 + num2;
        }

        // Method for subtraction
        public int Subtract()
        {
            return num1 - num2;
        }

        // Method to display values
        public void DisplayNumbers()
        {
            Console.WriteLine("First Number: " + num1);
            Console.WriteLine("Second Number: " + num2);
        }
    }

    class Program1
    {
        static void Main(string[] args)
        {
            Calculator calc = new Calculator();

            Console.Write("Enter first number: ");
            int a = Convert.ToInt32(Console.ReadLine());

            Console.Write("Enter second number: ");
            int b = Convert.ToInt32(Console.ReadLine());

            // Setting values
            calc.SetValues(a, b);

            // Display numbers
            calc.DisplayNumbers();

            // Perform operations
            Console.WriteLine("Addition = " + calc.Add());
            Console.WriteLine("Subtraction = " + calc.Subtract());
        }
    }
}
```

OUTPUT:

```
Enter First Number: 30
Enter Second Number: 20
Addition = 50
Subtraction = 10
|
```

CONSTRUCTOR:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDAY17
{
    internal class CalculatorConstructor
    {
        // Private data members
        private int num1;
        private int num2;

        // Constructor to initialize values
        public CalculatorConstructor(int a, int b)
        {
            num1 = a;
            num2 = b;
        }

        // Method for addition
        public void Add()
        {
            Console.WriteLine("Addition = " + (num1 + num2));
        }

        // Method for subtraction
        public void Subtract()
        {
        }
    }
}
```

```

        {
            Console.WriteLine("Subtraction = " + (num1 - num2));
        }
    }

class Program3
{
    static void Main(string[] args)
    {
        Console.Write("Enter First Number: ");
        int a = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Second Number: ");
        int b = Convert.ToInt32(Console.ReadLine());

        // Object creation with constructor
        CalculatorConstructor calc = new CalculatorConstructor(a, b);

        calc.Add();
        calc.Subtract();

        Console.ReadLine();
    }
}

```

OUTPUT:

```

Enter First Number: 50
Enter Second Number: 30
Addition = 80
Subtraction = 20
|

```